Supplementary Information: Fixed-Budget Local-Patch Risk Routing for Industrial Anomaly Detection

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1 Purpose and scope

This supplement records the unit definitions, paired resampling protocol, controlled baselines, failure cases, online-controller diagnostics, and latency measurement details supporting the main manuscript. All numerical values are from the strict matched-random evaluation unless explicitly labelled as an online or diagnostic result.

2 Evaluation units and uncertainty

An evaluation unit is one dataset category–random seed pair. The primary matched comparisons use 45 units for MVTec AD (15 categories $\times$ 3 seeds), 18 units for MPDD (6×3), and 36 units for VisA (12×3). For every unit, the local-routing score and the matched-random-selection score are computed with the same feature extractor, candidate patches, budget, seed, and downstream scoring rule. Reported effects are local minus matched-random values. Confidence intervals are percentile intervals from 10,000 paired bootstrap resamples of evaluation units. The sign-count column reports the number of units with a positive effect.

Table 1 Primary paired effects (local-patch risk minus matched-random), canonical 25% strong-routing result

Dataset	n	AUROC Δ	95% CI	+ units	Recall Δ	95% CI
MVTec AD	45	0.1249	[0.0837, 0.1659]	33/45	0.3877	[0.2916, 0.4792]
MPDD	18	0.1166	[0.0715, 0.1612]	16/18	0.1975	[0.1097, 0.2939]
VisA	36	0.0128	[0.0081, 0.0171]	28/36	0.2300	[0.1336, 0.3214]

The same 99 category–seed rows give a combined AUROC difference of +0.0826 (95% CI [0.0598, 0.1059]), a Recall@5%FPR difference of +0.2958 ([0.2347, 0.3551]), and 77/99 units with a positive AUROC effect. Against the additional routing baselines, local-patch risk exceeds the nearest-patch-distance dispersion controller by +0.0704 AUROC ([0.0446, 0.0990]) but only matches the fast-score controller (+0.0017, 95% CI [-0.0184, 0.0233]); on MPDD and VisA the fast-score controller is stronger.

3 Budget sensitivity

The MVTec budget sensitivity uses the same three-seed strict-quota protocol at 10%, 25% and 50% fallback budgets, with 45 category–seed units per budget. Table 2 reports the paired AUROC and Recall@5%FPR differences against the matched-random controller. The AUROC allocation benefit increases with budget, while the Recall benefit decreases; both remain strictly positive. These values describe allocation quality relative to matched-random routing and should not be interpreted as an end-to-end speedup.

Table 2 MVTec budget sensitivity (local-patch risk minus matched-random) at 10%, 25% and 50% fallback budgets

Budget	n	AUROC Δ	95% CI	Recall Δ	95% CI	+ units (AUROC / Recall)
10%	45	0.1145	[0.0821, 0.1484]	0.4981	[0.3940, 0.5977]	39/45, 42/45
25%	45	0.1249	[0.0837, 0.1659]	0.3877	[0.2916, 0.4792]	33/45, 42/45
50%	45	0.1589	[0.1226, 0.1954]	0.2479	[0.1834, 0.3110]	39/45, 35/45

4 Baseline and absolute-score context

On the 45 MVTec units, the full detector obtains AUROC 0.9392. At the primary 25% local budget, the strict local route obtains AUROC 0.8343 and the matched-random route 0.7094. These absolute values are not presented as superiority over the full detector; they establish that the contribution is allocation quality at a fixed reduced local budget. A PaDiM-style baseline obtains AUROC 0.8982 and

Recall@5%FPR 0.6544. A controlled PatchCore-like patch-level farthest-point coreset detector obtains AUROC 0.9433 and Recall@5%FPR 0.7847. The latter is an implementation-controlled approximation, not a claim of exact parity with the official PatchCore release.

5 Failure cases and boundary conditions

Three checks were retained as negative evidence. First, a multi-scale rank-fusion variant underperformed matched-random routing in a three-category smoke test, indicating that additional ranking signals can miscalibrate the budget allocation. Second, a lightweight cost filter underperformed the stronger patch-level coreset baseline. Third, the conformal-threshold diagnostic produced a realized fallback rate of 54.9% for a nominal 25% target and is therefore not used as a budget guarantee. These tests motivate the conservative claim that routing improves selection quality under a fixed experimental budget, not that every controller or every online policy is robust.

6 Online-prefix sensitivity

The online prefix controller was evaluated separately because it introduces a temporal prefix and fallback policy. On the 45 MVTec units, realized fallback was 25.34%, with AUROC effect 0.0331 (95% CI [0.0045, 0.0601]) and Recall effect 0.0804 (95% CI [0.0211, 0.1346]). The same controller produced AUROC/Recall effects of 0.0890/0.0643 on MPDD seed 17 and 0.00004/−0.0408 on VisA seed 17. With a VisA buffer of 128, the effects were 0.00124 (95% CI [−0.00113, 0.00339]) and 0.0750 (95% CI [0.0189, 0.1317]). These results are why online generalization is reported as configuration-sensitive rather than as a universal improvement.

7 Latency measurement

Latency was measured on the same machine with warm-up excluded from the reported percentile. Across 15 MVTec categories and 6,000 images with batch size one and CUDA synchronization, the Fast path had mean/P95 2.3292/2.9298 ms, the Full path 3.6269/4.0860 ms, and the Risk path (probe plus full) 5.1899/6.7302 ms. The Risk path includes both local and full operations and is a boundary diagnostic; it is not an end-to-end deployment benchmark. Batch throughput checks were limited to three categories and are treated as an extension rather than a primary claim.

8 Reproducibility record

The submission package records the random seeds, category lists, budget values, derived tables and figure source data. A submission-matched code release, including the environment specification and regeneration scripts, will be archived at https://github.com/cdhuangjin/fixed-budget-patch-routing. No private path is presented as a reproducibility link.